What Does a Model Expect?

Posterior Predictive Checks for Statistical Practice

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A model is a story about data

- We fit a model because it gives us a structured explanation of what we observed.
- That explanation also implies data sets that would be plausible under the model.
- A useful diagnostic asks: **does the model reproduce features we care about?**

The audience question

If the model generated a new data set, would it look recognizably like the one we saw?

Posterior predictive checks

After observing data y, simulate replicated data from

$$y^{\mathrm{rep}} \sim p(y^{\mathrm{rep}} \mid y) = \int p(y^{\mathrm{rep}} \mid \theta)\, p(\theta \mid y)\, d\theta.$$

1. Choose a feature or summary $T(y)$ that matters for the analysis.
2. Compare $T(y)$ with $T(y^{\mathrm{rep}})$ across replicated data sets.
3. Investigate a mismatch instead of treating it as a numerical failure.

What can a mismatch teach us?

A visible mismatch may indicate

- an omitted feature or dependence pattern;
- a distributional assumption that is too rigid;
- a scientific question the model was never built to answer.

But a match is limited

- the check is conditional on the chosen summary;
- a flexible model can look adequate by construction;
- diagnostics inform judgment rather than certify truth.

Takeaways

1. Posterior predictive checks make model assumptions concrete by simulating consequences.
2. The most useful checks connect directly to the scientific purpose of the analysis.
3. A good diagnostic changes what we ask next; it does not replace statistical judgment.

What would your model expect to see?